



BEST PRACTICE SERIES

Using VIMS to Manage Overcooled Engines

Maintenance and Repair

Application

Component Life Management

Component Renewal (CRC)

MARC
Management

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1.0 Introduction

Overcooled engines can lead to sludge formation in the crankcase. Sludge can gum up valve lifters and stems, as well as piston and piston ring. It can also cause corrosive wear and carbon buildup. Catastrophic failure could occur if these conditions went unnoticed.

The most common way engine overcooling can occur is for one or more of the jacket water thermostats to be stuck open. Other possible causes of engine overcooling are hydraulic fans remaining locked in the fully engaged condition, damage to the vent valve in the thermostat housing and extreme ambient temperatures.

A thermostat can become stuck open as a result of a cooling system that has been poorly maintained, through the use of raw water or inadequate levels of inhibitor/antifreeze. Raw water or poorly maintained coolant will allow the cast iron cylinder block to oxidize and deposit small iron particles into the coolant. These iron particles may lodge into the moving parts of the thermostats and cause them to stick open.

The recommended interval for thermostats to be changed is every 6,000 hours. However, there is a chance that a thermostat would fail between change-out intervals. Hence the situation must be frequently monitored in order to prevent catastrophic failure.

2.0 Best Practice Description

This Best Practice describes how to use VIMS to identify an overcooled engine. Plotting a histogram of coolant temperature is a great way to identify whether the truck is overcooled but the sample size must be big enough to get an accurate diagnosis. Additionally, datalogger can also assist in determining if the engine is overcooled.

Stuck open regulator

An engine temperature histogram can be used to identify overcooled engines. A normal histogram will show a tight temperature range with the majority falling within the target temperature range. Figure 2.1 shows a histogram of a 789B truck with normal engine coolant temperature. It indicates that 84% of samples have the temperature remaining between 80°C (176°F) and 85°C (185°F). There is only 7.65% below 80°C(176°F).

NOTE: Depending on the condition of the mine site, some trucks may have a thermostat set point of 90°C (194°F), but for demonstration purposes, the standard 80°C (176°F) thermostat is used.

Figure 2.2 shows the histogram of a 785B truck with irregular engine coolant temperature. There is no control of temperature (shown by the flat bell curve), and more than 20% of samples are below 80°C (176°F). This one has 91% of samples below 80°C (176°F). As a rule of thumb, a machine with any more than 20% samples below the thermostat's rating should be investigated for incorrect operation.

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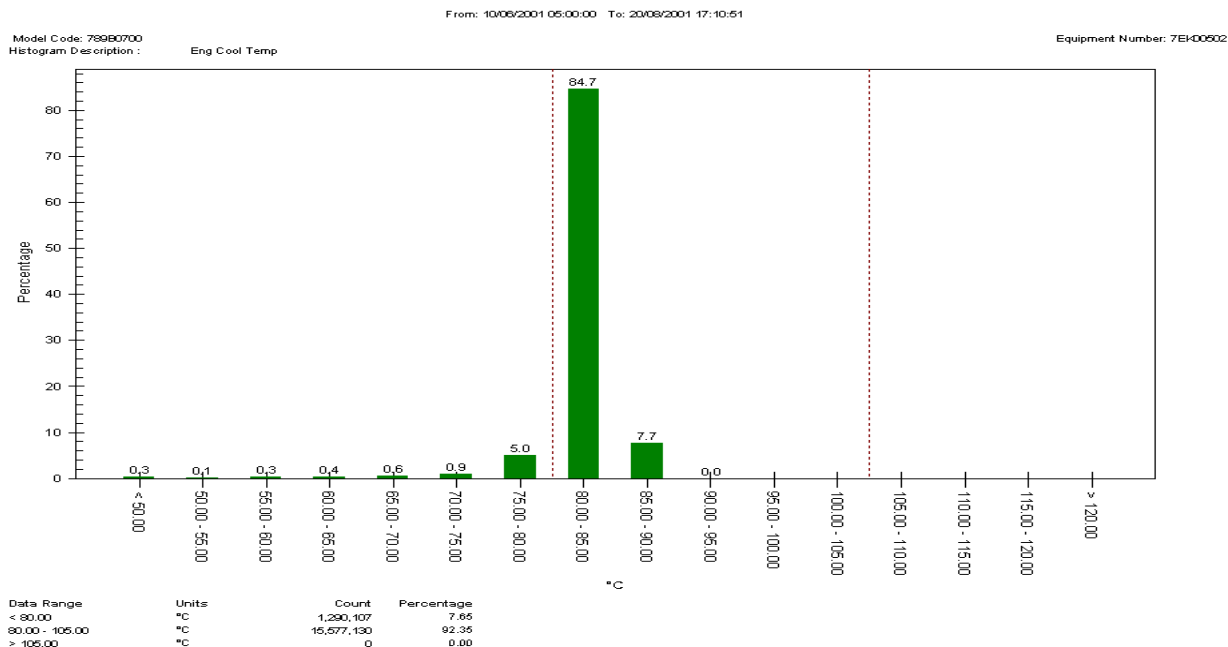


Figure 2.1: Histogram of the test 789B truck with normal engine coolant temperature

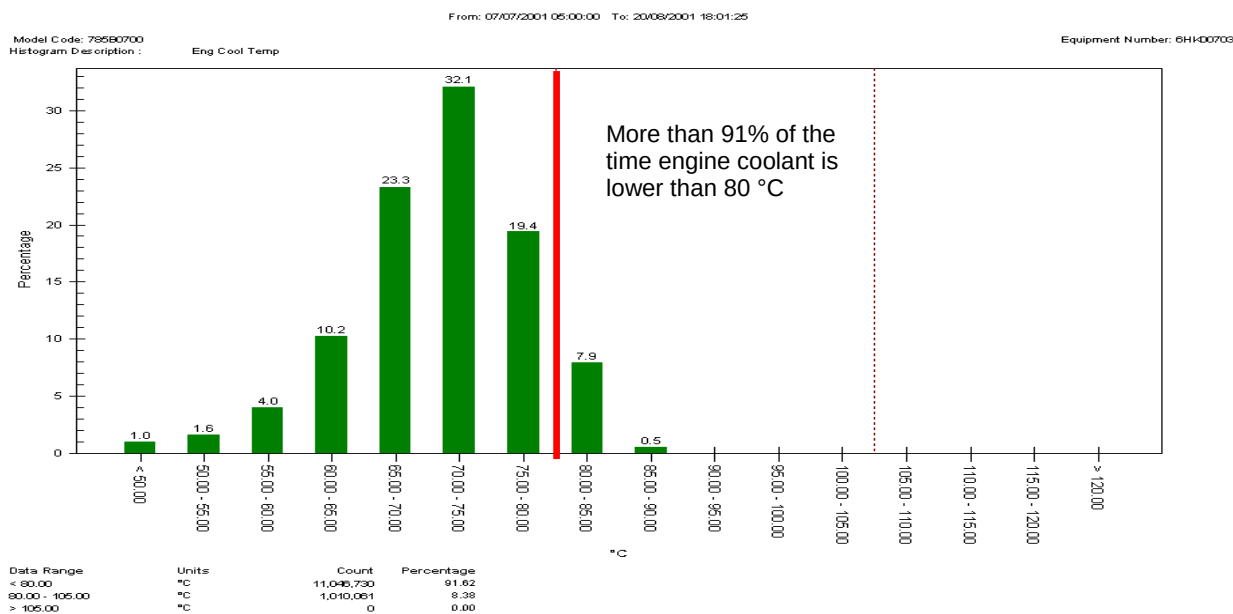


Figure 2.2: Histogram of the test 785B truck with irregular engine coolant temperature

Looking at the coolant temperature and Engine Load data from the datalogger can also identify an overcooled engine. A properly cooled engine will show coolant temperature near the set point when the engine is not under load, and higher coolant temperature when the engine is under load.

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Figure 2.3 shows the datalogger of a 789C truck with engine coolant temperature, payload status and load factor. Using normal conditions, the engine coolant temperature should remain at 80 – 82°C (176 – 179.6°F) when no load is applied. This is showed by the temperature “flat-lines” at the bottom graph.

Figure 2.4 shows the datalogger of a 785C truck. The temperature is below 80°C (176°F) all the time, even when the engine load factor is high. This, along with the engine temperature histogram (Figure 2.2) suggests the engine is being overcooled, which could be caused by the thermostat being stuck open. Action should be taken to investigate the cause of this situation.

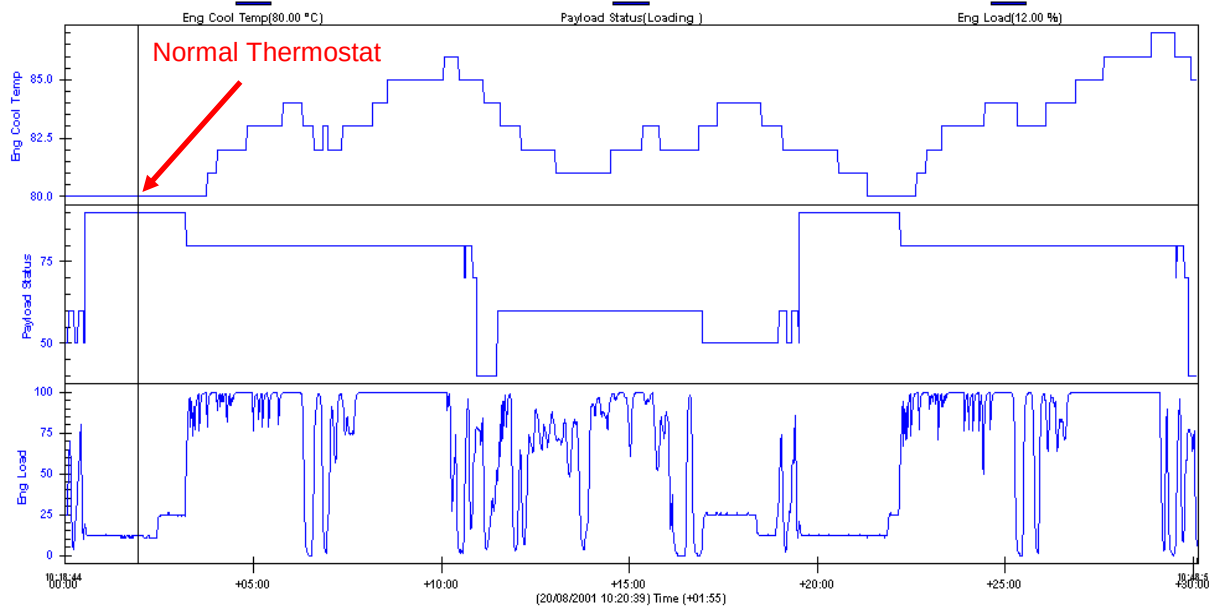


Figure 2.3: Datalog of the test 789C truck with normal engine coolant temperature

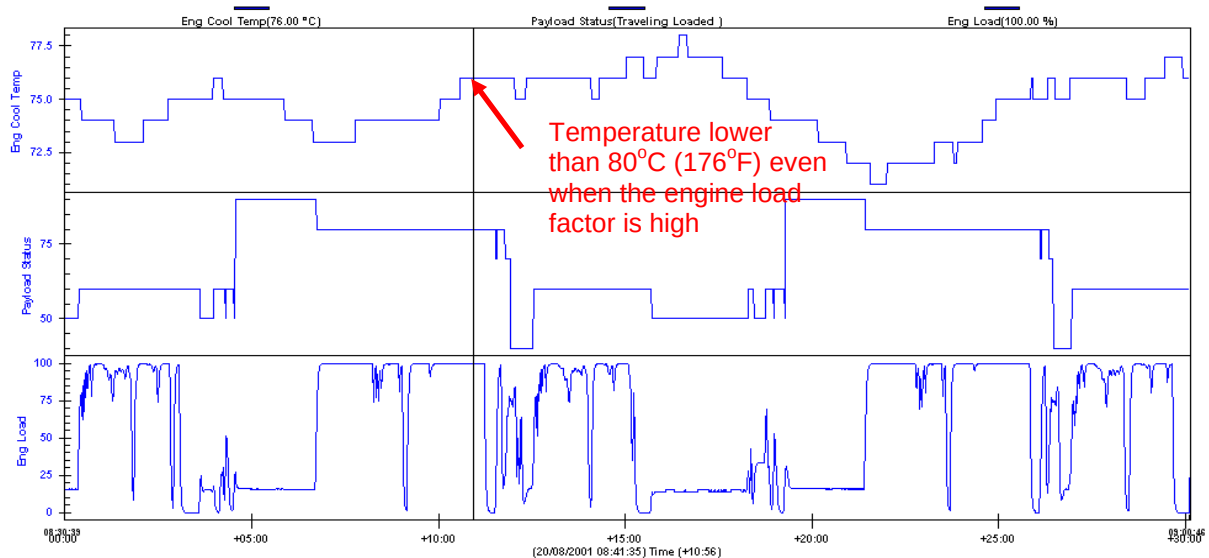


Figure 2.4: Datalog of the test 785B truck with irregular engine coolant temperature

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Sample Count for Histogram

Sample size is critical in the diagnosis process. If the sample size consists of the half a day of data, there is a possibility the truck spent most of its time idling. Data therefore won't present an accurate picture of machine health. A good sample size should include a large portion of time in normal operation. Usually, a week of data should give an accurate indication of whether the truck is overcooled or operating normally.

Damaged Vent Valve in Thermostat Housing

Another possible cause for low coolant temperature is a damaged check valve in the thermostat housing. A damaged valve can cause an alternate path for coolant to transfer back to the radiator. This coolant transfer is more prevalent in extremely cold conditions and can cause low coolant temperature. Operation in hot environments typically do not show significant effect on coolant temperatures when the vent valve is damaged.

When replacing thermostats the check valve should also be replaced to insure it is working properly.

Locked up Hydraulic Fan

Engine fan speed and coolant temperature can be used together to determine if the engine fan control and cooling circuit are working properly and not over cooling the engine. If the steering cycle blips can be seen in the engine fan speed graph or the engine coolant temperature drops below 88°C (190°F), the fan has likely remained fully engaged and is over cooling the engine.

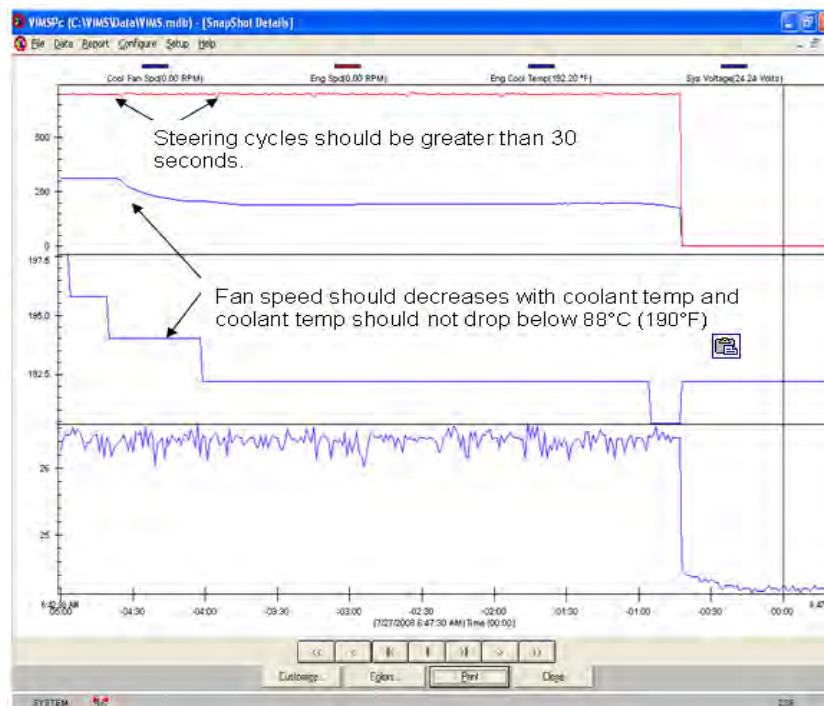


Figure 2.5: Datalogger data showing engine speed and fan speed data

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Extreme Ambient Air Temperature

If an application has significant periods of idle time and the ambient temperature is well below 0°C (32°F), even with a correctly operating thermostat, the engine jacket water temperature may drop below 79°C (175°F). Given enough time at cold enough temperature, the engine jacket water temperature may drop to 59°C (130°F) or lower. In these cases, fan clutches should be used to help keep the jacket water temperatures at or above 79°C (175°F). To minimize corrosive wear, Engine oil must be changed so the oil's TBN does not drop below 50% of new oil TBN (as per ASTM D2896). If the oil TBN gets too low it will not neutralize the acids that form during combustion. If these acids are not neutralized, corrosive wear will occur.

3.0 Implementation Steps

- Develop a process to regularly monitor engine temperature using VIMS data
- Assign personnel to monitor and track engine coolant temperatures, especially during cold seasons
- If low engine temperature is detected, the steps listed in this Best Practice should be used to identify and fix the problem

4.0 Benefits

The main benefits identified are:

- Prevent engine overcooling that might lead to catastrophic failure
- Reduced time in problem diagnosis
- Increased availability due to early diagnosis and prevention

5.0 Resources Required

- Fleet Analysis or Planner to analyze VIMS histogram, datalogger download and SOS sample at PM

6.0 Supporting Attachments / References

The Cause of Worn and Broken Piston Rings (SEPD0429-00)

Standard Test Method for Base Number of Petroleum Products (ASTM D2896)

VIMS Application Guide (SELD7003-03)

7.0 Related Best Practices

Using VIMS for Pre-PM Analysis (0409-1.01-1166)

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8.0 Acknowledgements

This Best Practice was written by:

Rex Cheung
Equipment Management Consultant
Caterpillar Global Mining
Cheung_Rex@cat.com
Ph: + 613 9953 9962

Our acknowledge to the personnel of Westrac for their willingness to share their idea and achievement that certainly will contribute to the success of other Mobile Equipment Maintenance Operation

Our special recognition to:

Adrian Clement, Westrac Equipment Support Group Manager
Adrian.Clement@westrac.com.au
Ph + 61 8 9377 9545

Steve De Boer, Westrac Reliability Engineer,
steve.deboer@westrac.com.au
Ph + 61 8 9377 9575

For further details and information regarding this Best Practice, please contact the author.

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